SCHOOL OF MATHEMATICAL AND COMPUTER SCIENCES
Actuarial Mathematics and Statistics

F78PA
Probability and Statistics A

Semester 1 - 2017/18
Duration: 2 hours

Question	Marks
1	18
2	13
3	21
4	13
5	9
6	11
Total Marks	85

Candidates must attempt ALL SIX questions.
To receive full credit you must show your work and explain your answers.

Approved electronic calculators may be used

AMS Formula Sheet and Cambridge Statistical Tables are provided

1. Consider the function

$$f_X(x) = \begin{cases} x, & \text{if } 0 \leq x \leq 1, \\ \frac{1}{2}, & \text{if } 1 < x \leq 2, \\ 0, & \text{otherwise.} \end{cases}$$

(a) Sketch the graph of the function f_X and explain why it is a density function for a random variable X. [3 marks]

(b) Determine ImX, the image of the random variable X and find $E(X)$, the expected value of X. [3 marks]

(c) Find F_X, the cumulative distribution function for X. [4 marks]

(d) Suppose that $Y = e^X$.

i) Explain why $ImY = [1, e^2]$ and show that for $1 \leq y \leq e^2$

$$F_Y(y) = F_X(\ln(y))$$

where F_Y is the cumulative distribution function of Y. [4 marks]

ii) Use part (d)(i) above to show that the density for Y is given by

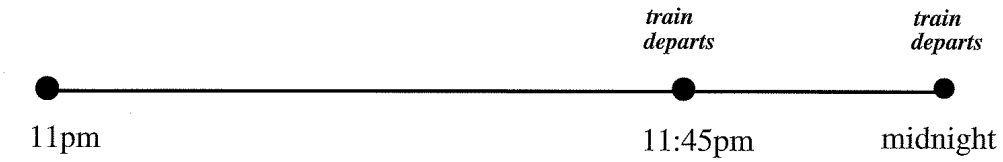
$$f_Y(y) = \begin{cases} \frac{\ln(y)}{y}, & \text{if } 1 < y \leq e, \\ \frac{1}{2y}, & \text{if } e < y \leq e^2, \\ 0, & \text{otherwise.} \end{cases}$$

[4 marks]

[Total 18 marks]

[PLEASE TURN OVER]

2. Fred arrives at a railway station at X minutes after 11.00 pm, where the random variable X is uniformly distributed on the interval $(0, 60)$ (i.e., $X \sim U(0, 60)$). Trains to Fred's destination depart at 11.45 pm, and midnight. Picture here:



(a) Let the random variable T be the length of time, in minutes, that Fred has to wait in the station for the next train to his destination.

i) Determine $\mathbf{P}(T \leq 10)$ and $\mathbf{P}(T \leq 30)$. [3 marks]

ii) Show that the distribution function, $F_T(t)$, for T is given by

$$F_T(t) = \begin{cases} 0, & t < 0, \\ \frac{t}{30} & 0 \leq t < 15, \\ \frac{1}{4} + \frac{t}{60} & 15 \leq t < 45, \\ 1 & 45 \leq t. \end{cases}$$

[4 marks]

iii) Determine $E(T)$. [3 marks]

(b) If Fred catches the train at 11.45pm, his train ticket will cost $£10$, but if he catches the train train at midnight, his ticket will cost $£15$.

Let Y denote the amount that Fred has to pay for his train ticket.

Determine $\mathbf{P}(Y = 10)$ and $\mathbf{P}(Y = 15)$ and compute $E(Y)$, the expected cost of Fred's train ticket. [3 marks]

[Total 13 marks]

3. Let X and Y be two random variables with joint probability density function

$$f_{X,Y}(x,y) = \begin{cases} 12y^2, & \text{for } 0 \leq x, y \text{ such that } 0 \leq x+y \leq 1, \\ 0, & \text{otherwise.} \end{cases}$$

(a) Sketch the region where the density $f_{X,Y}(x,y)$ is non-zero.

[1 mark]

(b) Show that the marginal density function $f_X(x)$ for X is given by

$$f_X(x) = \begin{cases} 4(1-x)^3, & 0 \leq x \leq 1, \\ 0, & \text{otherwise.} \end{cases}$$

[3 marks]

(c) **i)** Determine the conditional density function $f_{Y|X}(y|x)$ of Y given $X = x$. [2 marks]

ii) Find $\mathbf{P}(Y \leq 1/4 \mid X = 1/2)$ [3 marks]

iii) Find $E(Y \mid X = 1/2)$. [3 marks]

(d) Let $Z = X + Y$.

i) Explain why the image of Z is the interval [0,1] and show that the distribution function $F_Z(z)$ of Z is given by

$$F_Z(z) = \begin{cases} 0, & z < 0, \\ z^4, & 0 \leq z < 1, \\ 1, & 1 \leq z. \end{cases}$$

[6 marks]

ii) Calculate the conditional probability $\mathbf{P}(Z \leq 1/2 \mid Z > 1/4\ )$. [3 marks]

[$Total$ 21 marks]

[PLEASE TURN OVER]

4. Suppose that $X_1, X_2, \ldots, X_n$ are *independent* and *identically distributed* random variables with common distribution function $F_X(x)$ respectively. Let $M = \max\{X_1, X_2, \ldots, X_n\}$.

(a) Let $F_M(x)$ denote the distribution function for M. Show that for any $x \in \mathbb{R}$,

$$F_M(x) = (F_X(x))^n.$$

[4 marks]

(b) Anne and Bob are planning to paint a room. Anne will paint 2 walls and Bob will paint the other two walls.

Let X_A denote the time (in hours) that it takes Anne to paint two walls and let X_B denote the time (in hours) that it takes Bob to paint two walls and suppose that X_A and X_B are independent and identically distributed with common distribution function

$$F_X(x) = \begin{cases} 0, & x < 0 \\ x^2, & 0 \leq x < 1, \\ 1, & x \geq 1. \end{cases}$$

Suppose that Anne and Bob start painting at noon and let $T = \max\{X_A, X_B\}$ denote the time, in hours past noon, when Anne and Bob have finished painting the room.

i) Using part (a) above, show that the density $f_T(t)$ for T is given by

$$f_T(t) = \begin{cases} 4t^3, & \text{for } 0 < t < 1, \\ 0, & \text{otherwise.} \end{cases}$$

[3 marks]

ii) Let Y denote the time(in hours past noon) until the paint on the walls has dries.

Given that Anne and Bob finish painting the room at time $T = t$, the conditional distribution of Y is uniform between t+1 and t+2 (i.e. *given* that $T = t$, the conditional distribution of Y is $U(t+1, t+2)$)

Explain why

$$E(Y \mid T = t) = t + \frac{3}{2} \quad \text{for } 0 < t < 1.$$

[2 marks]

iii) Use the Partition Theorem, or otherwise, to compute $E(Y)$, the expected time (in hours past noon) until the paint has dried. [4 marks]

[Total 13 marks]

5. (a) Suppose X is a random variable with density given by

$$f(x) = e^{-x+1} \quad \text{for} \quad x \geq 1$$

(and $f(x) = 0$ otherwise).
Show that the moment generating function $M_X(\theta)$ for X is given by

$$M_X(\theta) = \frac{e^{\theta}}{1-\theta}$$

for $\theta < 1$.

Use $M_X(\theta)$ to determine $E(X)$, the expected value of X. [5 marks]

(b) Suppose that $Y = X - 1$ where X is the random variable described in part (a) and let $M_Y(\theta)$ denote the moment generating function for Y

Show that

$$M_Y(\theta) = \frac{1}{1-\theta}$$

and hence, or otherwise, determine the distribution of Y. [4 marks]

[$Total$ 9 marks]

6. The number of earth tremors detected in a week in a seismically active area of South America has a $Poisson(1)$ distribution, independent of the number of tremors detected in all other weeks.

Starting at $t = 0$, let X_1 denote the number of tremors detected in the first week of the year, let X_2 denote the number of tremors detected in the second week of the year, etc. and, in general, let X_n denote the number of tremors detected in the nth week of the year.

(a) Let $N = X_1 + X_2 + ... + X_{52}$ denote the total number of tremors detected in one year (i.e. 52 weeks). Use a moment generating argument to show that the *exact* distribution of N is $Poisson(52)$. (Be sure to give reasons) [5 marks]

(b) State carefully the Central Limit Theorem and use it to approximate the probability that, in total, more than 40 tremors are detected in one year (52 weeks), i.e. approximate $\mathbf{P}(N > 40)$. [6 marks]

[$Total$ 11 marks]

[END OF PAPER]